Deterministic Distance-Based Source Sampling for Efficient Multi-Scale Localised Network Centrality

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Abstract

Computing localised network centrality (closeness and betweenness) at multiple distance scales is computationally expensive at metropolitan scales, requiring several hours for a single multi-scale analysis. Source sampling reduces this cost, but comparative urban analysis requires a deterministic, network-agnostic schedule so that rankings remain directly comparable across locations.

We derive such a schedule $p(d)$ from the Hoeffding bound applied to a canonical grid reach model. Two parameters govern the bound: error tolerance ε (default 0.06), which controls the accuracy–speed trade-off, and failure probability δ (default 0.1). Validation on the Greater London (156,285 nodes) and Greater Madrid street networks at 1–20 km shows rank preservation ($\rho \geq 0.96$) for both closeness and betweenness, with speedups of up to 12.4×. The approach is implemented in the open-source `cityseer` Python package.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Urban morphological analysis relies on localised network centrality to reveal patterns of accessibility and flow in street networks (Cooper, 2015). Closeness and betweenness are computed within distance thresholds ranging from pedestrian (∼500 m) to metropolitan (∼20 km) scales, as implemented in tools such as sDNA (Cooper and Chiaradia, 2018). Computational cost is driven by the longest distances; for a network of 156,285 nodes, a multi-scale analysis spanning ten distance thresholds takes several hours.

When centrality patterns are compared within and between cities (Strano et al., 2013), the sampling schedule must be *deterministic* and *network-agnostic*: the same analysis distance must produce the same sampling probability regardless of network. If sampling rates varied by network, centrality values from different locations would reflect different noise levels, undermining cross-city comparison. Adaptive methods such as KADABRA (Borassi and Natale, 2019) provide convergence guarantees but produce network-dependent sample counts, precluding direct comparison.

Source sampling runs traversals from a random subset of source nodes and rescales the result. The mean number of reachable nodes (*reach*, r) governs both cost and accuracy: at short distances reach is small and traversals are cheap; at long distances reach is large and each sampled traversal updates many nodes, so both the potential speedup and the effective sample size per node grow with distance.

For global closeness, the Eppstein–Wang source-sampling bound (Eppstein and Wang, 2004) motivates replacing node count n with mean reach r, giving $k = O(\log r/\varepsilon^2)$. Path-sampling methods (Riondato and Kornaropoulos, 2016) are effective for global betweenness but lose their

cost advantage under localisation, where a single Brandes traversal (Brandes, 2001) updates all nodes in the bounded neighbourhood (Section 3). We therefore use a single deterministic source-sampling schedule $p(d)$ for both metrics.

Our contributions are:

1. A deterministic, network-agnostic source-sampling schedule $p(d)$ for localised centrality, obtained by (a) replacing global node count n with distance-bounded reach r in the Eppstein–Wang/Hoeffding bound, (b) substituting a canonical grid model for r so that p depends only on distance, and (c) applying a single schedule to both closeness and betweenness via shared Brandes traversals.

2. Empirical validation on the Greater London (156,285 nodes) and Greater Madrid networks at 1–20 km with $\varepsilon = 0.06$, demonstrating rank preservation ($\rho \geq 0.96$) with speedups of up to 12.4×.

2 Methods

We consider two centrality measures, both computed with a distance cutoff $d_{\max}$.

Harmonic closeness (Rochat, 2009; Freeman, 1979) sums inverse distances to all nodes within the threshold:

$$C_H(v;\, d_{\max}) = \sum_{\substack{u \neq v \\ d(v,u) \leq d_{\max}}} \frac{1}{d(v,u)} \tag{1}$$

Betweenness (Freeman, 1977) counts the fraction of shortest paths passing through each node. In the localised variant used here, each source s runs a Dijkstra bounded by $d_{\max}$; all targets t discovered by that traversal (i.e., with $d(s,t) \leq d_{\max}$) contribute to betweenness:

$$C_B(v;\, d_{\max}) = \sum_{\substack{s \neq t,\ s \neq v,\ t \neq v \\ d(s,t) \leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v. If v lies on a shortest path from s to t, then by subpath optimality $d(s,t) = d(s,v) + d(v,t)$, so the constraint $d(s,t) \leq d_{\max}$ implies $d(s,v) \leq d_{\max}$ and $d(v,t) \leq d_{\max}$; every node contributing to betweenness at threshold $d_{\max}$ is therefore discovered by the bounded Dijkstra.

Reach r is the mean number of nodes within $d_{\max}$:

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \leq d_{\max}\}| \tag{3}$$

We write d for the analysis distance $d_{\max}$ where the context is unambiguous (e.g., the sampling schedule $p(d)$). In a canonical grid network with inter-node spacing $s = 175$ m, reach grows from approximately 100 nodes at 1 km to over 40,000 at 20 km.

Synthetic networks. We generate synthetic networks using 3 topology templates: trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity). Each topology is tiled to create networks of approximately 12 km extent; we evaluate accuracy on a central 4 km×4 km window to reduce edge effects. We test 4 distance thresholds from 500 m to 4,000 m, and 5 error tolerances (ε) from 0.02 to 0.1.

Boundary effects. Nodes near network edges have systematically lower centrality because their catchments are truncated. We apply an inward buffer from the convex hull equal to the maximum analysis distance; only nodes inside this buffer are included in accuracy calculations. This follows standard practice for localised network analysis (Gil, 2017).

Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6\sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2-1)} \tag{4}$$

where R_i denotes the rank of node i. Urban applications rely on *relative* rankings (e.g., identifying the most central streets) rather than absolute values, so Spearman ρ measures the quantity of practical interest.

Normalised additive error $\varepsilon_{\text{obs}} = \max_v |C^{\text{sampled}}(v) - C^{\text{true}}(v)| \,/\, r$, where r is the canonical reach at the analysis distance. This normalisation places closeness and betweenness errors on a common per-node scale for diagnostic comparison. (The Hoeffding bound itself uses metric-specific upper bounds $C_{\max}$; see Section 3.)

Validation networks. We validate on two real-world street networks: the Greater London Authority (GLA) road network (Ordnance Survey Open Roads; $\approx$156,285 nodes after preprocessing) and the Greater Madrid metropolitan network. Both use a 20 km inward buffer to eliminate edge artefacts.

3 A Unified Sampling Model and Deterministic Distance Schedule

On synthetic networks, Spearman ρ exceeds 0.95 even at ε values well above the default of 0.06 for both closeness and betweenness (Figure 1). The tree and linear topologies exceed $\rho = 0.95$ at $\varepsilon = 0.06$ by a wide margin. The trellis topology falls slightly below the threshold, particularly for betweenness. Its regular grid structure produces many nodes with near-identical centrality, so small perturbations cause rank swaps that depress ρ even though the spatial pattern of centrality is preserved. Among the three topologies tested, the trellis represents the worst case for rank-based evaluation; the mean across topologies remains above 0.95.

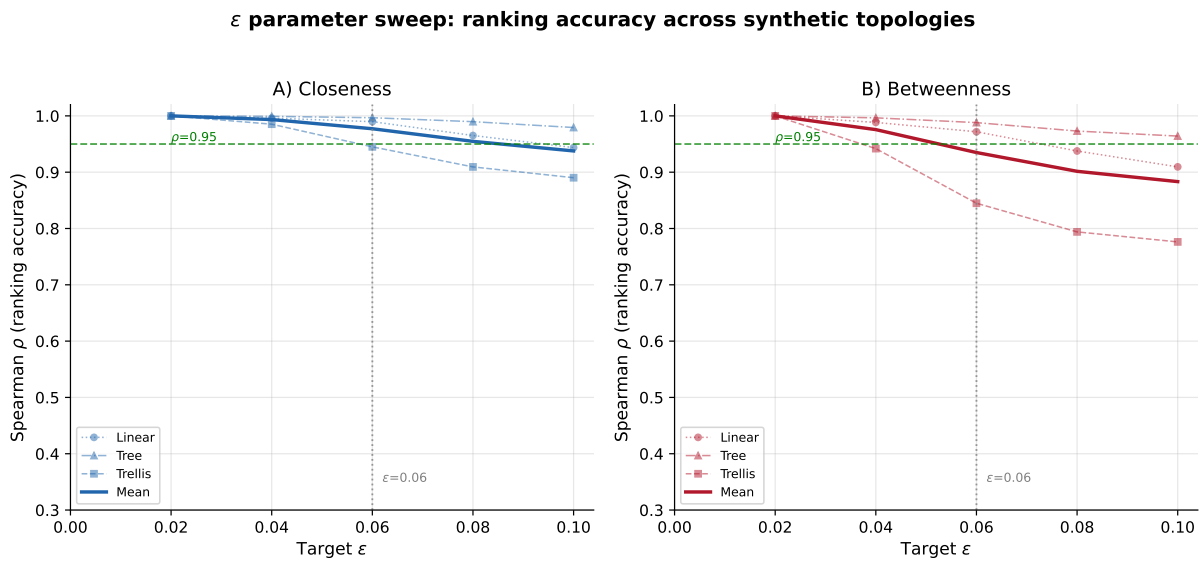

Figure 1: **The sampling opportunity.** Ranking accuracy (Spearman ρ) vs. error tolerance ε for closeness (A) and betweenness (B), across synthetic topologies and distances. Per-topology lines are shown alongside the bold mean. The green dashed line marks $\rho = 0.95$; the grey dotted line marks the paper default $\varepsilon = 0.06$. The trellis topology falls below the threshold because its regular grid produces many near-identical centrality values, making rank ordering sensitive to small perturbations (see text).

Closeness: Hoeffding source-sampling. Following Eppstein and Wang (Eppstein and Wang, 2004), each of k uniformly sampled sources contributes a value in $[0, 1]$ (after normalising by $C_{\max}$) to each target node's closeness estimate. The Hoeffding bound (Hoeffding, 1963) with a union bound over target nodes yields sample complexity $k = \log(2n/\delta)/(2\varepsilon^2)$ for global centrality. Under localisation with threshold $d_{\max}$, only nodes within $d_{\max}$ of a given target contribute to its centrality estimate. The union bound therefore applies over at most r nodes (the mean neighbourhood size) rather than all n nodes in the network. Substituting r for n gives

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2} \tag{5}$$

with corresponding sampling probability

$$p = \min\left(1,\ \frac{k}{r}\right) \tag{6}$$

This model has no fitted parameters: ε (error tolerance) and δ (failure probability) are user-chosen constants. At default values $\varepsilon = 0.06$, $\delta = 0.1$:

- **Low reach** ($r \lesssim 1{,}600$): $k > r$, so $p = 1$ and computation is exact.
- **High reach**: p decreases as $O(\log r/r)$, so speedup $1/p$ grows as $O(r/\log r)$.

From reach-based to deterministic sampling probability. Equations 5–6 express p as a function of reach r, which varies across networks. To obtain a network-agnostic schedule (Section 1), we substitute a canonical grid model: $r \approx \pi d^2/s^2$ with spacing $s = 175\,\mathrm{m}$, chosen to represent sparse urban morphology (typical European block lengths range from 80–180 m). Most real urban networks are denser than this, so the canonical reach underestimates actual reach and the prescribed p oversamples relative to what any given network requires. The schedule trades some potential speedup for robust rank preservation across heterogeneous morphologies.

Betweenness: source-sampling versus path-sampling. Existing betweenness approximation methods fall into two families.

Source-sampling selects a subset of source nodes and runs a full Brandes traversal (Dijkstra plus backpropagation) from each. Each traversal updates dependency scores for all r nodes in the bounded neighbourhood simultaneously. Brandes and Pich (Brandes and Pich, 2007) established this approach empirically, scaling the result by n/k (where k is the number of sampled sources); Geisberger et al. (Geisberger et al., 2008) improved it with distance-weighted corrections to reduce overestimation near the source; Bader et al. (Bader et al., 2007) used adaptive source-sampling for per-vertex estimation. None of these methods provides a deterministic, distance-based sampling schedule.

Path-sampling (Riondato and Kornaropoulos, 2016; Borassi and Natale, 2019; Pellegrina and Vandin, 2023) instead samples random source–target pairs and computes a single shortest path per sample, updating betweenness counts only along that path. Sample complexity depends on the vertex diameter rather than on n, yielding formal (ε, δ) guarantees:

$$m = O\left(\frac{\log \mathrm{VD} + \log(1/\delta)}{\varepsilon^2}\right) \tag{7}$$

However, under a localisation cutoff $d_{\max}$, the admissible pair set depends on local morphology (no fixed global pair space), making path-sampling difficult to adapt to distance-bounded analysis. Moreover, each path sample updates only one path, whereas a single source traversal updates all r reached nodes; and path-sampling applies only to betweenness, so closeness would require a separate computation.

Because each source traversal already runs a bounded Dijkstra, betweenness can be estimated from the same samples used for closeness with no additional traversal cost; the Dijkstra exploration is shared, and backpropagation adds $O(r)$ work per source. A single bounded Dijkstra traversal from source s with Brandes backpropagation accumulates dependency contributions[1] $\delta_s(v)$ for all r reached nodes. Each $\delta_s(v)$ is bounded in $[0, r-1]$, and maximum betweenness within the neighbourhood is at most $r(r-1)$. After normalising by $C_{\max}$, the per-source contribution lies in $[0, 1/r]$, and the same Hoeffding argument yields sample complexity identical to the closeness case (Equations 5–6).

The Hoeffding bound is nevertheless more conservative for betweenness than for closeness, for two reasons: dependency values across target nodes are not independent (they share a shortest-path tree), and betweenness contributions are more variable than closeness contributions, making the range-based bound less tight. This is consistent with the empirically thinner margins observed for betweenness (Section 4).

Table 1 provides the deterministic sampling probabilities at standard analysis distances for several ε values.

Table 1: Deterministic sampling schedule at standard analysis distances under different error tolerances ($\delta = 0.1$, $s = 175\,\mathrm{m}$). Paper default: $\varepsilon = 0.06$ for both metrics.

Distance	$\varepsilon = 0.02$		$\varepsilon = 0.04$		$\varepsilon = 0.06$		$\varepsilon = 0.08$		$\varepsilon = 0.1$	
	k	p	k	p	k	p	k	p	k	p
500 m	exact		exact		exact		exact		exact	
1 km	exact		exact		exact		exact		exact	
2 km	exact		exact		exact		exact		exact	
5 km	exact		exact		1,506	58.7%	847	33.0%	542	21.1%
10 km	exact		3,822	37.3%	1,698	16.6%	955	9.3%	611	6.0%
20 km	17,022	41.5%	4,255	10.4%	1,891	4.6%	1,063	2.6%	680	1.7%

Error structure across reachability. Figure 2 shows a crossover: absolute error increases with reach, but relative error (absolute error divided by median true centrality) decreases. High-reach nodes have higher effective sample sizes, so their relative errors are small; *precision scales with reachability.*

Relationship to prior bounds. Table 2 summarises existing sampling methods, grouped by computational primitive; Bergamini et al. (2019) provide a comprehensive empirical comparison of these approaches for global betweenness.

The schedule at $\varepsilon = 0.06$ is empirically conservative: observed $\varepsilon_{\mathrm{obs}}$ values on both validation networks are well below ε, and relative errors decrease with reach (Figure 2B). Three sources of slack account for this: (i) the Hoeffding bound uses only the range, ignoring the smaller actual variance; (ii) the union bound does not exploit positive dependence among node-level errors; (iii) Spearman ρ is more forgiving than additive error, since small absolute errors rarely alter rankings among well-separated nodes. Bernstein's inequality (Boucheron et al., 2013) would tighten source (i) by incorporating empirical variance, which on the synthetic networks tested here is small relative to the range ($\sigma^2/b^2 \approx 0.02$); however, the Hoeffding bound is already conservative enough that this refinement is unnecessary for the accuracy targets considered here.

Each distance threshold draws an independent source set S_d. In the implementation, distances where $p_d = 1$ are batched into a single pass (one Dijkstra per source serves all such distances simultaneously), while each sampled distance ($p_d < 1$) requires a separate pass with

[1] Brandes' notation $\delta_s(v)$; not to be confused with the failure probability δ.

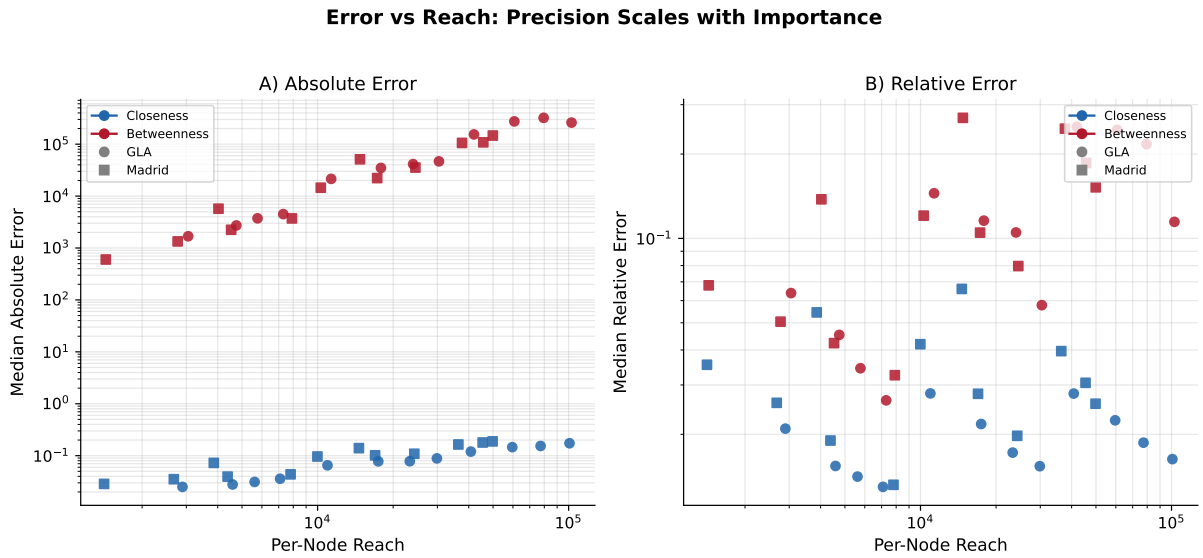

Figure 2: **Error crossover: precision scales with reachability.** Per-node median errors pooled across reach quartiles at distances where $p < 1$, for Greater London (circles) and Greater Madrid (squares). (A) Absolute error increases with reach. (B) Relative error (absolute error divided by median true centrality value) decreases with reach.

Table 2: Sampling approaches for centrality approximation. Methods are grouped by computational primitive: *source-sampling* runs a full Brandes traversal from each sampled node, updating all reached nodes; *path-sampling* samples individual shortest paths, updating only nodes along that path. All prior methods target global centrality; our approach replaces node count n with distance-bounded reach r and applies one schedule to both metrics.

Method	**Primitive**	**Metric**	**Guarantee**	**Complexity**	**Key param.**
Source-sampling (full Dijkstra tree per sample)					
Eppstein & Wang (2004)	Source	Closeness	Additive, all nodes	$O(\log n/\varepsilon^2)$	n
Brandes & Pich (2007)	Source	Both	Empirical	User-chosen k	—
Bader et al. (2007)	Source	Betweenness	Relative, per vertex	$O(1/\varepsilon^2 b(v))$	$b(v)$
Geisberger et al. (2008)	Source	Betweenness	Unbiased estimator	User-chosen k	—
Path-sampling (single shortest path per sample)					
Riondato & Korn. (2016)	Path	Betweenness	Additive, all nodes	$O(\log \mathrm{VD}/\varepsilon^2)$	VD
KADABRA (2019)	Path	Betweenness	Additive, all nodes	Adaptive	D
SILVAN (2023)	Path	Betweenness	Non-uniform	Variance-dep.	Variance
This work	**Source**	**Both**	**Additive, all nodes**	$\boldsymbol{O(\log r/\varepsilon^2)}$	$\boldsymbol{r}$

Algorithm 1 Deterministic Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Error tolerance ε (default 0.06), confidence δ (default 0.1)
Require: Canonical grid spacing s (default 175 m)
Ensure: Closeness estimates $\hat{C}_H[d]$ and betweenness estimates $\hat{C}_B[d]$ for each $d \in D$

```
for each distance d ∈ D do
    Compute canonical reach: r_d ← πd²/s²
    Compute required sources: k_d ← log(2r_d/δ)/(2ε²)
    Compute sampling probability: p_d ← min(1, k_d/r_d)
    Sample sources S_d ⊆ V by including each node independently with probability p_d
    for each source s ∈ S_d do
        Run distance-bounded Dijkstra from s; accumulate closeness via inverse distances
        Run Brandes backpropagation; accumulate betweenness path counts
    end for
    Scale both estimates by 1/p_d                    ▷ Horvitz–Thompson; Appendix A
end for
return {Ĉ_H[d], Ĉ_B[d] : d ∈ D}
```

its own source set. A node sampled at one distance is not reused at another; per-distance independence simplifies the probabilistic analysis at the cost of some redundant traversals.

4 Validation

We evaluate on the GLA and Greater Madrid networks using Algorithm 1 with $\varepsilon = 0.06$ and $\delta = 0.1$, reporting the mean over three random seeds. Ground truth uses all sources ($p = 1.0$).

Greater London. Table 3 reports results at five distances from 1–20 km. Both metrics achieve $\rho > 0.95$ at all distances, with minimum $\rho = 0.9672$ (betweenness, 20 km). Wall-clock speedup reaches 12.4× at 20 km.

Table 3: Sampling validation on Greater London network ($\varepsilon = 0.06$, $\delta = 0.1$, $s = 175$ m).

Dist.	p	ρ_c	**Spd**$_c$	ρ_b	**Spd**$_b$
1 km	100.0%	1.0000	0.5×	1.0000	0.5×
2 km	100.0%	1.0000	0.5×	1.0000	0.5×
5 km	58.7%	0.9957	0.9×	0.9972	0.8×
10 km	16.6%	0.9938	3.1×	0.9857	3.5×
20 km	4.6%	0.9932	12.4×	0.9672	12.2×

Network: Greater London Area. Deterministic distance-based schedule: same p for both metrics at each distance. Subscripts: c = closeness, b = betweenness.

At distances where $p = 1.0$ (1–2 km), the measured speedup falls below 1.0× because the sampling machinery adds overhead without reducing the number of traversals. Wall-clock speedup exceeding 1.0× emerges at 10 km and beyond. Betweenness is consistently the binding constraint ($\rho_{\text{betw}} < \rho_{\text{close}}$ at every distance), consistent with its heavier-tailed contribution distribution (Section 3).

Greater Madrid. Madrid's metropolitan area includes sparser peripheral zones where actual reach falls below the canonical curve (Figure 5), so the schedule provides less oversampling than for London. The schedule still achieves $\rho > 0.95$ for both metrics at all tested distances (Table 4), though the margin is thinnest at 20 km betweenness ($\rho = 0.9600$), the weakest result across both networks. Wall-clock speedup reaches 9.9× at 20 km.

Table 4: Sampling validation on Greater Madrid network ($\varepsilon = 0.06$, $\delta = 0.1$, $s = 175\,\mathrm{m}$).

Dist.	p	ρ_c	**Spd**$_c$	ρ_b	**Spd**$_b$
1 km	100.0%	1.0000	0.5×	1.0000	0.2×
2 km	100.0%	1.0000	0.5×	1.0000	0.6×
5 km	58.7%	0.9966	0.8×	0.9971	0.8×
10 km	16.6%	0.9914	3.0×	0.9848	3.1×
20 km	4.6%	0.9819	9.9×	0.9600	13.1×

Network: Greater Madrid, 64,609 nodes. Deterministic distance-based schedule: same p for both metrics at each distance.

Comparison. Together, the two networks bracket a range of European urban morphologies: London is denser, with actual reach consistently above the canonical curve, so the schedule oversamples and accuracy margins are wide; Madrid's sparser peripheral zones bring actual reach closer to or below the canonical estimate, providing a harder test of the schedule's robustness. That the schedule achieves $\rho \geq 0.95$ on both networks, with betweenness at 20 km as the binding case in each, is consistent with the canonical grid model serving as a conservative lower bound across this morphological range (Figure 3).

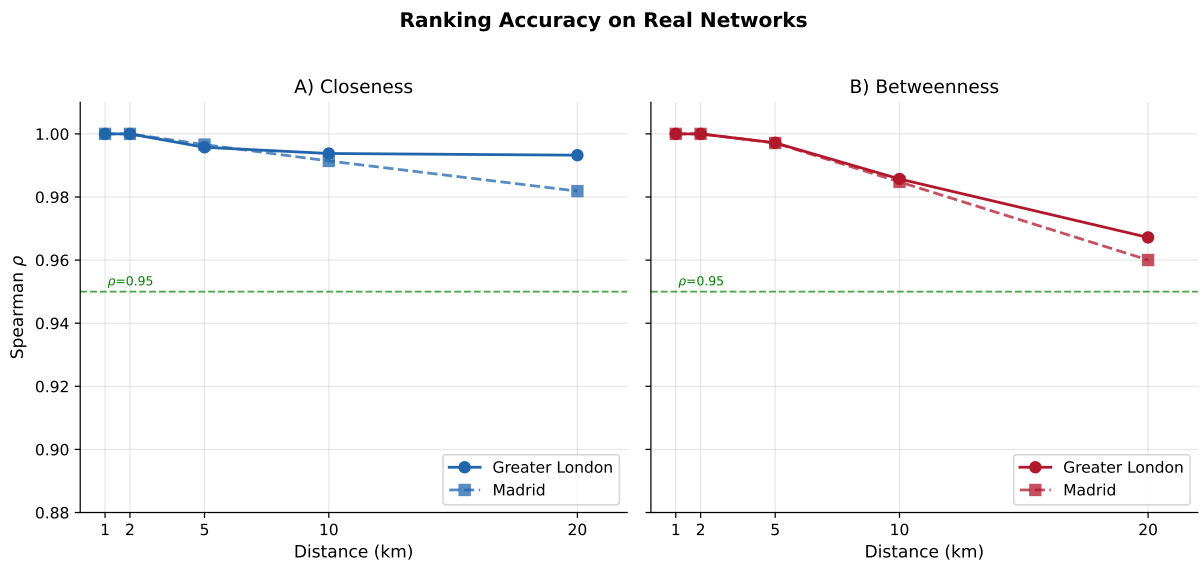

Figure 3: **Ranking accuracy on real networks.** Spearman ρ vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). The green dashed line marks $\rho = 0.95$. Both networks exceed the target at all distances; betweenness is the binding constraint.

5 Discussion and Conclusion

Practical guidance. In `cityseer`, the user sets ε and the library derives $p(d)$ for each distance threshold, applying it to both metrics. For manual implementations, Algorithm 1 gives the complete procedure. Table 5 and Figure 6 provide the deterministic schedule at standard analysis distances.

At the default $\varepsilon = 0.06$:

- $\leq$2 km: $p = 1.0$ (exact; no speedup).

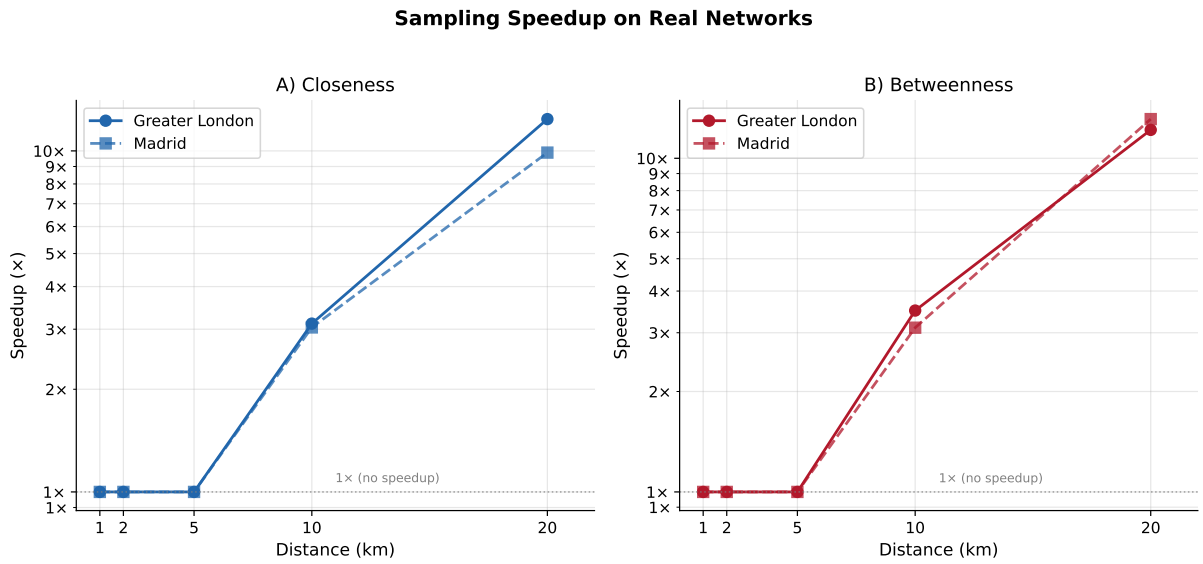

Figure 4: **Measured wall-clock speedup on real networks.** Speedup (baseline time / sampled time, log scale) vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). At short distances where p is close to 1, sampling overhead dominates and no wall-clock speedup is observed; speedup exceeding 1.0× emerges at 10 km and beyond. At long distances, GLA achieves higher measured speedup partly because the exact baseline traversals are costlier on the denser network; the deterministic schedule remains conservative for GLA (Figure 5).

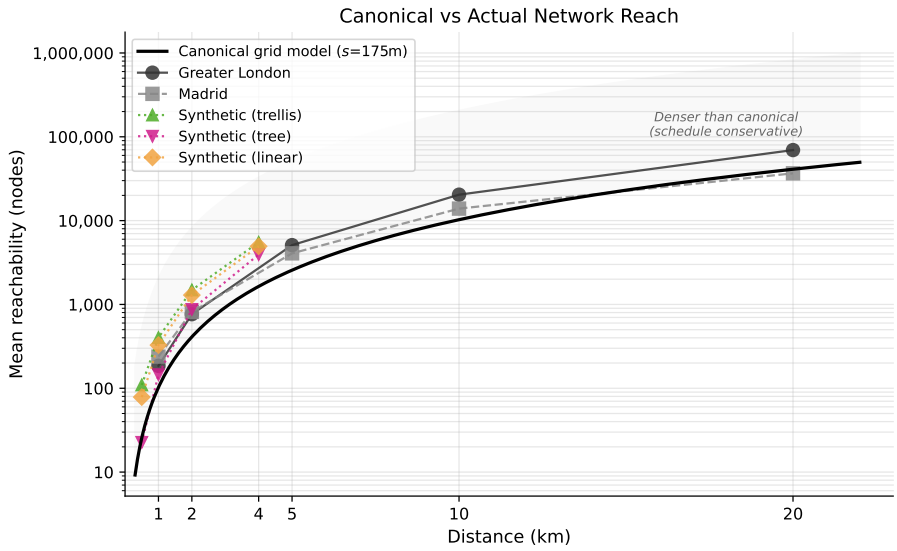

Figure 5: **Canonical vs. actual network reach.** Mean reach (log scale) vs. analysis distance for Greater London, Greater Madrid, and three synthetic topologies, compared against the canonical grid model ($s = 175$ m, black line). Greater London lies consistently above the canonical curve (the schedule oversamples). Greater Madrid falls below at longer distances, reflecting sparser peripheral areas. Both validation networks achieve $\rho \geq 0.95$ at all distances (Tables 3–4).

Table 5: Deterministic sampling schedule by analysis distance ($\varepsilon = 0.06$, $\delta = 0.1$, canonical grid spacing $s = 175\,\mathrm{m}$). Both closeness and betweenness use the same schedule. Sampling is exact ($p = 100\%$) for distances up to 4 km. Speedup shown is theoretical ($1/p$); measured wall-clock speedup is lower at near-unity p due to sampling overhead and emerges at 10 km and beyond.

Distance	**Canonical reach**	k	p (%)	**Speedup** ($1/p$)
1 km	103	1,059	100	1×
2 km	410	1,252	100	1×
5 km	2,565	1,506	58.7	1.7×
10 km	10,258	1,699	16.6	6.0×
20 km	41,033	1,891	4.6	21.7×

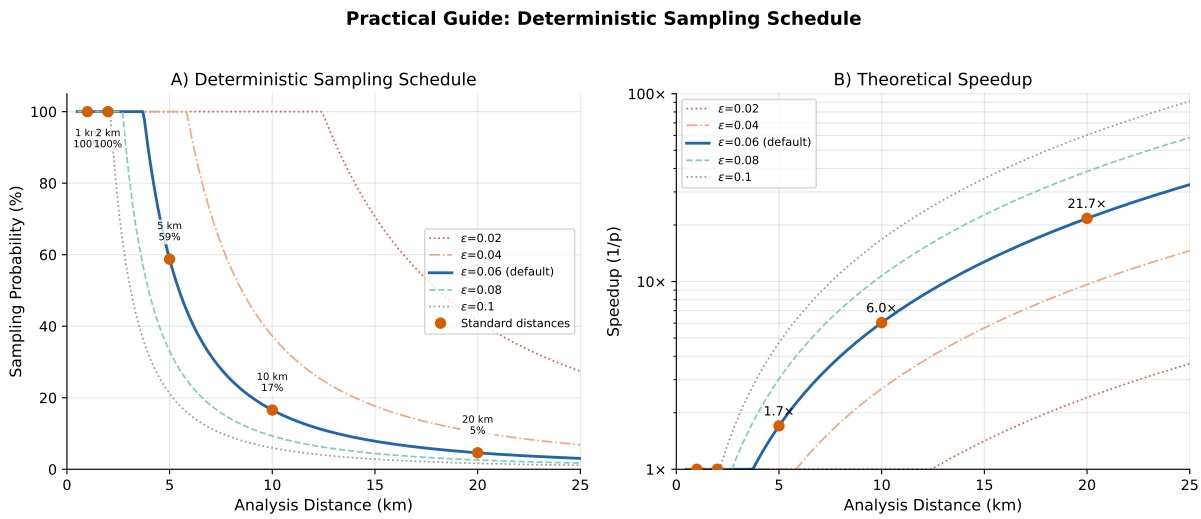

Figure 6: **Practical guidance.** (A) Deterministic sampling probability p vs. analysis distance under the canonical grid model ($\varepsilon = 0.06$, $\delta = 0.1$), with standard analysis distances annotated. (B) Theoretical speedup $1/p$ vs. distance; measured wall-clock speedup is lower at near-unity p due to sampling overhead (see Figure 4).

- 5 km: $p \approx 59\%$ (1.7× theoretical speedup).
- 10 km: $p \approx 17\%$ (6×).
- 20 km: $p \approx 5\%$ (∼12.4× measured).

Because $k \propto 1/\varepsilon^2$, halving ε quadruples the required sources. In practice: $\varepsilon = 0.04$ eliminates sampling below 10 km but halves the speedup at 20 km (Table 1); $\varepsilon = 0.06$ (default) maintains $\rho \geq 0.96$ on both validation networks; $\varepsilon = 0.1$ still yields $\rho > 0.99$ on synthetic networks (Figure 1). Betweenness is the binding constraint on the schedule; users computing only closeness may use $\varepsilon = 0.08$ or 0.1 while maintaining $\rho > 0.95$ (Figure 1A).

For small networks (< 1,000 nodes), exact computation completes in seconds and sampling offers no practical benefit. For untested morphologies, a pilot comparison at one representative long distance (e.g., 10 km) with and without sampling can verify the schedule: $\rho > 0.95$ indicates sufficient rank preservation for that morphology; if not, reducing ε increases accuracy.

Limitations. First, we assume uniform random sampling of source nodes. Non-uniform strategies (e.g., stratification or importance weighting) could achieve higher accuracy with fewer samples, but require proxies for node importance that are unavailable before computation.

Second, substituting canonical grid reach for actual network reach means denser networks are oversampled while sparser networks rely on the Hoeffding slack for accuracy (as observed for Madrid; Figure 5). Networks with inter-node spacing substantially exceeding 175 m (e.g., rural road networks) would warrant separate validation.

Third, the model treats the network as statistically homogeneous: accuracy depends on distance rather than local structure. Validation on the GLA and Madrid networks, which include dense urban cores and sparse suburbs, suggests the schedule is conservative overall, but accuracy may vary locally.

Fourth, validation uses shortest-path (metric) distance only. Angular distance (simplest-path), as used in space syntax (Hillier and Hanson, 1984; Hillier et al., 1993) and road-centre-line analysis (Turner, 2007), produces different reachability patterns and may require different ε calibration; this warrants separate validation.

Fifth, results are averaged over three random seeds (with small inter-seed variation in ρ), and validation covers two European cities; other morphologies would strengthen external validity.

Future work. Adaptive per-node sampling (varying p with local reach) could improve efficiency in heterogeneous networks. Online stopping criteria (Borassi and Natale, 2019) could replace a priori calibration but require new theory for localised centrality. Variance-dependent bounds (Pellegrina and Vandin, 2023) offer a path to tighter schedules on well-structured networks. Extending validation to angular (simplest-path) distance would be necessary before applying the schedule in space syntax research.

Comparative urban analysis requires that sampled centrality values be directly comparable across networks. We achieve this by replacing network-specific node count n with distance-bounded reach r in the Eppstein–Wang/Hoeffding bound, then substituting a canonical grid model so that the sampling schedule $p(d)$ depends only on analysis distance. The resulting schedule serves both closeness and betweenness from shared Brandes traversals, and its error structure concentrates precision on high-centrality nodes while low-reach peripheral nodes approach exact computation.

Validation on the Greater London and Greater Madrid street networks at 1–20 km confirms rank preservation ($\rho \geq 0.96$) for both metrics, with wall-clock speedups of up to 12.4×. The canonical grid model intentionally overestimates the required sample count for most urban morphologies, trading maximum speedup for robustness; networks sparser than the canonical spacing may warrant separate validation. Within these bounds, the schedule makes multi-scale centrality analysis at metropolitan scales practical without compromising the cross-network comparability on which urban morphological research depends.

The model is implemented in the open-source `cityseer` Python library.[2] Analysis scripts and validation data are available in the repository's `analysis/sampling/` directory.

References

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A Technical Details

Synthetic network generation. Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12 km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

GLA network preprocessing. The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that do not represent junctions)
5. Remove dangling nodes (dead-end segments $< 20\,\mathrm{m}$)
6. Consolidate nearby nodes (threshold $1\,\mathrm{m}$)
7. Extract largest connected component

Final network: $\approx$ 156,285 nodes.

Correctness verification against NetworkX. We verified the Rust-based `cityseer` implementation against NetworkX (Hagberg et al., 2008) across all three synthetic topologies, distance thresholds from $200\,\mathrm{m}$ to $3200\,\mathrm{m}$, both distance heuristics, and both metrics. The maximum per-node difference

$$\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)| \tag{8}$$

was $\max_v \Delta_v < 10^{-6}$ in all configurations, attributable to floating-point precision.

Horvitz–Thompson estimator. The $1/p$ rescaling used in Algorithm 1 is a Horvitz–Thompson (H–T) estimator (Horvitz and Thompson, 1952): under Bernoulli source sampling with inclusion probability $\pi_i = p$, the population total is estimated as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where S is the set of sampled sources. This inverse-probability weighting is unbiased whenever $\pi_i > 0$.

For closeness, the per-source contribution $y_s(v) = 1/d(s, v)$ is determined solely by source s, so contributions are independent across sources and unbiasedness follows directly. For betweenness, although the metric involves source–target *pairs*, the Brandes decomposition expresses betweenness as $C_B(v) = \sum_s \delta_s(v)$, where $\delta_s(v)$ aggregates all of source s's pair-dependency contributions to v via backpropagation. Because each $\delta_s(v)$ depends only on the shortest-path tree rooted at s, the per-source terms are independent under Bernoulli sampling, and the H–T estimator $\hat{C}_B(v) = \sum_{s \in S} \delta_s(v)/p$ is unbiased by the same argument.

An alternative is the Hájek ratio estimator (Hájek, 1971),

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i}, \tag{9}$$

which is often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). We use Horvitz–Thompson because it extends directly to unequal inclusion probabilities, should future work combine base sampling probability with per-source importance weights.

Implementation in `cityseer`. The sampling model is implemented in the `cityseer` Python library (version 4.23+). Sampling is enabled via the `sample=True` flag. The `node_centrality_shortest` function computes both closeness and betweenness from shared Dijkstra traversals in a single call:

```
from cityseer.metrics import networks

gdf = networks.node_centrality_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
    sample=True
)
```

Convenience wrappers (`closeness_shortest`, `betweenness_shortest`) are available for computing a single metric, but using them separately forgoes the shared-traversal advantage.

The default error tolerance ($\varepsilon = 0.06$) and confidence ($\delta = 0.1$) are set in `cityseer.config`. Helper functions for custom workflows are available in `cityseer.config`, including:

- `compute_hoeffding_p`: Hoeffding bound from mean reach
- `compute_distance_p`: deterministic p from distance via canonical grid model

Reproducibility. All analysis scripts and validation results are available at analysis/sampling/. The current script pipeline runs sequentially from `00_generate_cache.py` through `05_generate_macros.py` via `run_all.py`, with each script's outputs documented in its header.